

Complex Conjugate Pairs

Key Points

A complex number, $z = x + yi$ has a complex **conjugate**

$$z^* = x - yi$$

Alternative notation for the conjugate is $\bar{z}, \hat{z}$

$$\text{We know } zz^* = x^2 + y^2 (= |z|^2)$$

When a quadratic equation has a complex solution the other solution is the conjugate

Basic Skills

Q1. Write down the complex conjugate of the following

- a) $z = 5 + i$
- b) $z = 6 - 2i$
- c) $z = 2(3i - 1)$

Q2. Solve the following quadratic equations

- a) $z^2 + 4z + 5 = 0$
- b) $z^2 + 9 = 0$
- c) $z^2 - 2z + 5 = 0$

Competency Skills

Q3. Solve the following quadratic equations

- a) $2z^2 + z + 5 = 0$
- b) $4z^2 = -16$
- c) $8z + 5(1 - z^2) = 0$
- d) $2z = \frac{z^2}{4} + 25$

Q4. Given that a quadratic equation has the solution $z = 3 + 4i$ write other solution and hence find the equation which has these solutions

Q5. Given that a quadratic equation has the solution $z = 5 - i$ write other solution and hence find the equation which has these solutions

Advanced Skills

Q6. Given that a solution to the cubic $z^3 - 5z^2 + z + 75$ is $z = 4 - 3i$ find the other solutions

Q7. Given that a factor of the cubic $z^3 - 5z^2 + 9z - 5$ is $(z - i + 2)$, fully factorise $z^3 - 5z^2 + 9z - 5$

Q8. Calculate zz^* for the following

- a) $z = 4 + 2i$
- b) $z = 8 - 3i$

Q9. In general for $z = x + yi$ prove the expression for $z\bar{z} = x^2 + y^2$

Q10. Explain why the complex solutions always appear in conjugate pairs, using algebraic reasoning

Extension

Q11. Given that a quartic, $az^4 + bz^3 + cz^2 + dz + e$, has two solutions $z = 2 + 3i$ and $z = -5 + i$ find a, b, c, d and e